User Defined Commands

1 New Commands - Static

BibTeX
Typesetting BibTeX with the new command.

Some logos: TeX LaTeX

Here is an equation $a^2 = b^2 + c^2$, more $a^2 = b^2 + c^2$

2 New Commands - Dynamic

This is the *output* of sentence. This is the second output.
This is the ***output*** of sentence. This is the second again output.

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena. Necessity depends on, when thus treated as the practical employment of the never-ending regress in the series of empirical conditions, time. Human reason depends on our sense perceptions, by means of analytic unity. There can be no doubt that the objects in space and time are what first give rise to human reason.

Let us suppose that the noumena have nothing to do with necessity, since knowledge of the Categories is a posteriori. Hume tells us that the transcendental unity of apperception can not take account of the discipline of natural reason, by means of analytic unity. As is proven in the ontological manuals, it is obvious that the transcendental unity of apperception proves the validity of the Antinomies; what we have alone been able to show is that, our understanding depends on the Categories. It remains a mystery why the Ideal stands in need of reason. It must not be supposed that our faculties have lying before them, in the case of the Ideal, the Antinomies; so, the transcendental aesthetic is just as necessary as our experience. By means of the Ideal, our sense perceptions are by their very nature contradictory.

Trying the `\texcmd` to produce `\LaTeX`

```
\newcommand{\cmd}{def}
```

If $y = f(u)$ and $u = f(x)$, then $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$
Integrating x: $\int_0^{\inf} x\,dt$
Differentiating x by time: $\frac{dx}{dt}$

2.1 Embedding Optional Arguments

2.1.1 One as Optional

Default
New

2.1.2 More than one, 1st being Optional

def/den three
new/den three again
This is a Red Text Vs. New Red Text

2.2 Enhancing/Integrating User Commands

The x-command is now printing x Moreover, it's value can be integrated inside the new $x\,y$

3 New Commands with Two Optional Arguments

Mandatory with Default Optional A, B, Third Argument

Overriding First Optional Only First New, B, Third Argument

Overriding both Optional 1st, 2nd, Still Three

Empty First Argument , Second, Three Only

Empty Second Argument First, , Third

Empty both Optional , , Three Only

3.1 Renewing cmds with two optional args

<u>Q</u> P *Some input text*

P *Some input text*

<u>`Q`</u> *Some input text*

3.2 Omitting Defaults of Optional Arguments

Some text

First Only Third Mandatory

4 Redefining Commands

4.1 Altering the Built-in Commands

A

Tabular i: Cell Table

Some **New Text bf** surrounding text.

4.1.1 Be Careful Redefining Existing cmds

The `\vec` command was $\vec{v}$ but now it's $\mathbf{v}$
So, we need to have a new command to exhibit this $\vec{\boldsymbol{v}}$

4.2 Altering the User-defined Commands

The `\xy` is now printing xyz

5 Nesting Definitions of New Commands

5.1 Using \newcommand

Father contains Son Content and Son Content
Father contains Son Content and Son Content
Father contains Son Content and Son Content

Second Call To call the outer command more than once you have to surround the inner commands with {}.

Scope of Inner If the inner commands are surrounded by {}, then you cannot use them outside their scope, but if no brackets exist, then you can use them freely globally in the document.

5.2 Using \providecommand

Father contains Son Content and Son Content

Outer contains Son Content and Grand Content
Outer contains Son Content and Grand Content

Provide Behavior Allows for modular and conditional definitions of commands in LaTeX. It's a feature often used in package development or when writing complex documents to ensure that commands are available when needed, but also to prevent errors due to command name clashes.

Scope of Inner Note that the `\Inner` is accessible outside the parent container, Son Content and also Grand Content

5.3 Arguments for Nested Definitions

Father contains Son Content and Grand Content and also Fourth Level !